Vijay Kumar Chauhan

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Research Profile

Battery researcher specializing in experimental characterization, heat-generation analysis, battery safety, electrochemical-thermal modeling, machine learning, battery management systems and thermal management. Research integrates high-precision calorimetry with physics-based and data-driven approaches to develop experimentally validated models for lithium-ion batteries under realistic operating conditions. Current research extends these approaches toward battery Digital Twins, predictive thermal management and fault/anomaly detection.

Research Interests

Lithium-Ion Batteries • Battery Thermal Management Systems (BTMS) • Battery Safety and Thermal Runaway • Electrochemical-Thermal Modeling • Heat Generation and Calorimetry • Battery Management Systems (BMS) • State-of-Charge Estimation • Equivalent Circuit Modeling (ECM) • Machine Learning for Batteries • Battery Digital Twins • Predictive Thermal Management • Fault and Anomaly Detection • Fast Charging • Battery Aging

Education

Direct Ph.D., Sustainable Energy Engineering *IIT Kanpur, India (2021–Present)*
Specialization: Battery Thermal Management Systems
CGPA: 7.6 / 10
Thesis submitted; Ph.D. defense/viva expected in December 2026

B.Tech., Mechanical Engineering *SITM, Greater Noida, Uttar Pradesh, India (2017)*
Percentage: 80%

Research Experience

Ph.D. Research Scholar, IIT Kanpur *Aug 2021 – Present*
Research: Experimental Battery Characterization, Battery Safety, Electrochemical-Thermal Modeling, Machine Learning, BMS and BTMS

- Conducted extensive experimental investigations on lithium-ion batteries using isothermal and adiabatic calorimetry to quantify heat generation, thermal behaviour and battery safety characteristics.
- Performed comprehensive battery safety and abuse testing on lithium-ion cells ranging from 1 Ah to 100 Ah in cylindrical, pouch and prismatic formats.
- Conducted thermal abuse experiments using an adiabatic rate calorimeter (ARC), including controlled heating toward self-heating onset and thermal runaway, to characterize thermal stability and heat-release behaviour.
- Performed electrical abuse testing including overcharge, over-discharge and external short-circuit tests to investigate electrical safety limits and failure behaviour.
- Performed mechanical abuse testing including nail penetration and crush testing to investigate internal short-circuit formation, thermal runaway initiation and cell failure mechanisms.
- Experimentally measured heat generation rate (HGR) of LFP, NMC and LCO lithium-ion cells over a wide range of temperatures and discharge rates using an isothermal battery calorimeter.
- Developed experimentally informed heat-generation correlations using machine learning and symbolic regression for prediction of battery HGR from operating conditions.
- Developed coupled electrochemical-thermal models based on porous-electrode theory and heat-transfer modeling in COMSOL Multiphysics for predicting voltage, heat generation and temperature under realistic EV drive cycles including FUDS and US06.
- Performed electrochemical parameter estimation, sensitivity analysis and model optimization using Python, SciPy and PyBaMM.
- Developed compact analytical HGR correlations using Python-based symbolic regression (PySR) and machine-learning models including FNN, LSTM, Random Forest, K-NN and Support Vector Machine.

- Investigated temperature, C-rate and SOC dependence of battery heat generation and identified operating conditions associated with efficient battery operation and thermal behaviour.
- Developed lithium-ion battery State-of-Charge estimation methods including Coulomb Counting, EKF, AKF and Hybrid AKF-LUT approaches incorporating OCV-based correction.
- Investigated voltage hysteresis, polarization and their influence on battery heat generation and thermal response during charging and discharging.
- Developed and characterized passive battery thermal management concepts incorporating phase-change materials (PCM) and natural circulation cooling for battery modules.
- Currently developing a battery Digital Twin integrating experimentally derived HGR correlations, electrochemical-thermal models and battery operating conditions to predict internal heat generation and subsequent cell thermal behaviour.
- Currently developing a predictive/adaptive BTMS framework in which predicted heat generation is used as an early thermal-load indicator to anticipate cell temperature evolution and enable proactive thermal-management actions.
- Investigating fault and anomaly detection approaches by comparing measured battery behaviour with model-predicted electrical and thermal responses.
- Estimated thermophysical properties including specific heat capacity and anisotropic thermal conductivity of lithium-ion cells.
- Fabricated coin cells and performed electrochemical characterization including EIS and cycling experiments.
- Mentored more than 75 undergraduate and postgraduate students in battery testing, experimentation and laboratory activities.

Selected Publications

1. **V.K. Chauhan**, J. Bhattacharya, "Error estimation of Bernardi heat evaluation of Li-ion cells under drive cycle operation," **Applied Thermal Engineering**, 254 (2024). **[SCI, IF=7.5]** DOI.

2. **V.K. Chauhan**, M. Sahu, J. Bhattacharya, "Thermal characterization of 18650 lithium iron phosphate cell for wide ranges of temperature and discharge rate to identify most efficient operating window," **Journal of Energy Storage**, 119 (2025) 116274. **[SCI, IF=10.7]** DOI.

3. **V.K. Chauhan**, J. Bhattacharya, "Developing an accurate correlation for estimating heat generation rate by an 18650 lithium iron phosphate cell through a machine learning model trained on experimental measurements," **Energy**, 334 (2025). **[SCI, IF=10.1]** DOI.

4. **V.K. Chauhan**, M. Sahu, J. Bhattacharya, "Comparing heat generation rate and application suitability of 18650 $LiFePO_4$, Li(Ni-Mn-Co)O_2 and $LiCoO_2$ cells under diverse conditions," **Applied Thermal Engineering**, 298 (2026) 131051. **[SCI, IF=7.5]** DOI.

5. **V.K. Chauhan**, J. Bhattacharya, "A multistage data-driven method for deriving a compact heat-generation correlation from high-precision calorimetry data of lithium-ion batteries: A case study for 18650 NMC cell," **International Journal of Thermal Sciences**, 229 (2026) 111055. **[SCI, IF=5]** DOI.

6. **V.K. Chauhan**, J. Bhattacharya, "High-Fidelity SOC Estimation of Lithium-Ion Batteries Using Isothermal Calorimetry and a Hybrid Adaptive Kalman Filter with OCV-Based Correction," **Under submission to Journal of Energy Storage**.

Conferences and Workshops

1. **V.K. Chauhan**, J. Bhattacharya, *Estimation of Heat Generation in 18650 $LiCoO_2$ Cells under Realistic Drive Cycles Using P2D Numerical Model and Analytical Bernardi Formulation*, Proceedings of the 2nd International Meeting on Energy Storage Devices (IMESD-2023) & Industry–Academia Conclave, IIT Roorkee, 2023.

2. **V.K. Chauhan**, J. Bhattacharya, *Investigation of Heat Generation and Temperature Evolution for Cylindrical Cells under Actual Drive Cycle Operation: A Combined Approach through Experiment and Simulation*, Proceedings of the 1st International Conference on Advances in Sustainable Solutions for Energy Transitions (ASSET-2025), IIT Guwahati, 2025.

Technical Expertise

Battery Research: Lithium-Ion Batteries, LFP, NMC, LCO, HGR Measurement, Battery Calorimetry, Thermal Characterization, Thermal Runaway, Battery Safety and Abuse Testing
Modeling: Electrochemical-Thermal Modeling, P2D Modeling, ECM, 2RC Models, Thermal Modeling, Parameter Estimation, Hysteresis Modeling
BMS/BTMS: SOC Estimation, Coulomb Counting, EKF, AKF, Hybrid AKF-LUT, Battery Thermal Management, Predictive Thermal Management
Machine Learning: Python, TensorFlow/Keras, Scikit-learn, PySR, LSTM, FNN, Random Forest, K-NN, SVM, Symbolic Regression
Simulation: COMSOL Multiphysics, PyBaMM, ANSYS Fluent, MATLAB/Simulink
Experimental: ISO-BTC+, BTC-500+ Adiabatic Calorimeter, ARC, Thermal Abuse Testing, Electrical Abuse Testing, Mechanical Abuse Testing, Nail Penetration, Crush Testing, Overcharge, Over-discharge, External Short Circuit, Internal Short Circuit, EIS, Coin Cell Fabrication, NI-DAQ

Certifications and Courses

- **Introduction to Battery-Management Systems**, University of Colorado Boulder, Coursera, 2023.
- **Equivalent Circuit Cell Model Simulation**, University of Colorado Boulder, Coursera, 2024.
- **Supervised Machine Learning: Regression and Classification**, DeepLearning.AI and Stanford University, Coursera, 2024.
- **Advanced Learning Algorithms**, DeepLearning.AI and Stanford University, Coursera, 2024.
- **Programming for Everybody (Getting Started with Python)**, University of Michigan, Coursera, 2023.
- **MATLAB Simulink Onramp & Optimization Onramp**, MathWorks.

Professional Activities

- **Reviewer:** Journal of Energy Storage (Elsevier).
- **Reviewer:** Applied Thermal Engineering (Elsevier).
- **Organizer:** Research Scholar Day (RSD), Department of Sustainable Energy Engineering, IIT Kanpur, 2024.

Achievements

- Received research fellowship from the Ministry of Education, Government of India, for Ph.D. research at IIT Kanpur.
- Qualified Graduate Aptitude Test in Engineering (GATE) in Mechanical Engineering three times.

References

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Prof. Vaibhav Arghode
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